OncoRepurpose AI

A Clinical Decision Support & Research App for Drug Repurposing in Oncology

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🧠 App Overview

OncoRepurpose AI is a mobile application that helps clinicians, researchers, and students identify **repurposed drugs for cancer treatment** by integrating:

- Drug databases
- Molecular pathways
- AI-based recommendations
- **Molecular docking (binding affinity)**

👉 The goal: **faster, evidence-based, and mechanistically supported cancer therapy insights**

🎯 Core Objectives

- Discover **new uses for existing drugs**
- Provide **mechanism-based explanations**
- Support **personalized oncology decisions**
- Integrate **bioinformatics + structural biology**

⚙️ FULL APP FEATURES (Integrated)

🔍 1. Cancer-Based Search Engine

Users enter:

- Cancer type (e.g., breast, lung)

📊 Output:

- List of repurposed drugs:
  - Propranolol
  - Metformin

💊 2. Drug Intelligence Database

Each drug includes:

📌 Basic Info:

- Original use
- Repurposed oncology role

🧬 Mechanism of Action:

- Example:
  - Propranolol → inhibits β-adrenergic signaling
  - Metformin → activates AMPK → inhibits mTOR

🧬 3. Pathway & Gene Analysis

Displays:

🎯 Target Genes:

- VEGF → angiogenesis
- BAX / BCL2 → apoptosis

🔄 Pathways:

- Angiogenesis
- Apoptosis
- mTOR signaling

👉 Helps explain *why the drug works*

📊 4. Evidence Level Indicator

Each drug is labeled:

- 🟢 Clinical trials
- 🟡 In vitro / in vivo
- 🔵 Computational

👉 Adds scientific credibility

🤖 5. AI Recommendation System

📄 Input:

- Gene expression profile
  (e.g., VEGF ↑, BAX ↓)

⚡ Output:

- Suggested drugs:
  - Propranolol

🧠 Logic (simple rule-based):

- VEGF ↑ → anti-angiogenic drugs
- mTOR ↑ → Metformin

🧬 6. Molecular Docking & Binding Affinity Module (KEY FEATURE)

🎯 Purpose:

Validate drug–protein interaction

🧪 Example Display:

Drug: Propranolol
Target Protein: VEGFR

Binding Affinity (ΔG):
👉 **-7.5 kcal/mol**

📊 Interpretation:

- 🟢 Strong binding (≤ -7)
- 🟡 Moderate (-5 to -7)
- 🔴 Weak (≥ -5)

🔬 Interaction Details:

- Hydrogen bonds
- Hydrophobic interactions

💡 What this adds:

👉 Structural validation of drug action
👉 Makes your app **research-level**

📦 APP STRUCTURE (USER FLOW)

🏠 1. Home Screen

- Search cancer type
- Quick buttons:
  - "AI Recommendation"
  - "Explore Drugs"

📄 2. Results Screen

- List of repurposed drugs

- Filters:
    - Evidence level
    - Binding strength

🔬 3. Drug Details Screen

Tabs:

📌 Overview

- Drug info

🧬 Mechanism

- Pathways + genes

📊 Evidence

- Study level

🧲 Docking

- Binding affinity
- Protein target
- Interaction summary

🤖 4. AI Recommendation Screen

- Input gene markers
- Output:
    - Suggested drugs
        -
            - docking strength

🛠 TECHNICAL IMPLEMENTATION (Simple & Realistic)

📱 Frontend:

- Flutter / React Native

🗄 Backend:

- Firebase / local database

🤖 AI:

- Rule-based system (no ML required)

📊 Example Logic:

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IF VEGF ↑ AND docking score ≤ -7
→ Recommend Propranolol

IF mTOR ↑
→ Recommend Metformin
```

👩‍⚕️ TARGET USERS

- Oncologists
- Pharmacists
- Researchers
- Biotechnology students

🚀 FUTURE IMPROVEMENTS

- Real-time integration with databases (TCGA, GEO)
- Machine learning prediction
- Clinical trial matching
- Nanoparticle drug delivery suggestions

🏆 FINAL PROJECT DESCRIPTION (Ready to Submit)

“OncoRepurpose AI is a mobile application that integrates drug repurposing, molecular pathway analysis, and ligand–protein binding affinity data to provide evidence-based and mechanistically supported recommendations for cancer therapy.”